

```

const int wakeupPin = 4; // GPIO 4 for external wake-up
RTC_DATA_ATTR int bootCount = 0; // Number of reboots

int pHSense = 35; // p0
int samples = 10;
float adc_resolution = 4095.0; // ESP 32 12 bit ADC Resolution
int measurings = 0;

float ph(float voltage)
{
    return 7 + ((2.50 - voltage) / 0.18);
}

void setup()
{
    Serial.begin(115200); // Start serial port at 115200 baud rate
    pinMode(wakeupPin, INPUT_PULLUP); // Declare touch sensor pin as INPUT_PULLUP
    bootCount++; // Increment the number of boots by 1
    Serial.println("Boot number: " + String(bootCount)); // Print the boot number at execution

    for (int i = 0; i < samples; i++)
    {
        measurings += analogRead(pHSense);
        delay(10);
    }
    float voltage = 3.3 / adc_resolution * measurings / samples;
    Serial.print("pH= ");
    Serial.println(ph(voltage));
    delay(1000);
    esp_sleep_enable_ext0_wakeup((gpio_num_t)wakeupPin, HIGH); // Configure ext
    wakeup delay(1000); // Adding a 1 second delay to avoid multiple presses
    Serial.println("I'm going to sleep now."); // Print the statement before entering deep
    sleep esp_deep_sleep_start(); // Enter deep sleep mode
}

void loop()
{
    // We will never use loop
}

```

Fig. 5: Snippet of the source code

Specific nutrients necessitate certain pH levels for optimal plant uptake. Incorrect pH levels can lead to deficiencies in certain nutrients or toxicity. For instance, when the pH level drops below 5.0, plants may develop magnesium and calcium deficiencies, or copper and iron toxicity. Fig. 3 illustrates a typical pH scale and its correlation to everyday objects in our lives.

ESP32 power modes

Active mode. In this mode, all peripherals of the chip remain active, including Wi-Fi, Bluetooth, and radio based on program input.

Modem-sleep mode. The CPU is operational, and the clock is configurable. The Wi-Fi/Bluetooth baseband and radio are disabled.

Light-sleep mode. The CPU is paused, while the RTC memory, RTC peripherals, and the ULP coprocessor are running. Wake-up events (MAC, SDIO host, RTC timer, or external interrupts) will awaken the chip.

Deep-sleep mode. Only the RTC memory and RTC peripherals are powered up. Wi-Fi and Bluetooth connection data are stored in the RTC memory. The ULP coprocessor remains functional.

Hibernation mode. The internal 8MHz oscillator and ULP co-processor are disabled. The RTC recovery



Fig. 6: Additional information about various aspects of the setup